

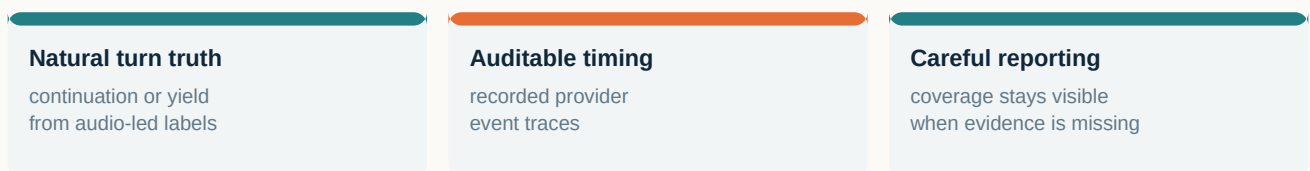
ASLI / TURNBENCH

TurnBench

A disciplined way to test whether a voice agent has spoken too soon.

TurnBench is a separate, reusable offline benchmark. It measures a simple but important moment in a conversation: after a speaker pauses, did the agent answer before that speaker actually resumed?

It is designed to make turn-taking claims auditable. The current v1 ships with local fixtures and strict reporting rules, not a live provider ranking or a cross-language performance claim.

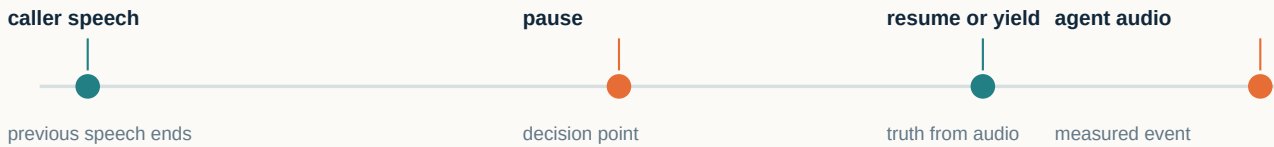
**OFFLINE V1**

Released with ASLI on 21 August 2026. Source of truth: the versioned local JSONL records, the scoring code, and the annotation guide in the repository.

01 / THE PROBLEM

A pause is evidence. It is not a verdict.

A pause can be a hesitation inside the same thought, or it can be a real handoff. A voice agent must not confuse the two.



The benchmark asks: did the agent speak before the caller resumed?

The question TurnBench answers

For each labelled decision after a pause: did first audible agent audio happen before the caller resumed, or how long after a true yield did the caller wait?

WHAT IT TESTS	WHAT IT DOES NOT ASSUME
Turn boundary behaviour on matched audio, labels, and event traces.	That one filler word, punctuation mark, transcript ending, or silence threshold tells us a caller is finished.
The earliest audible agent reply, not merely a backend decision signal.	That missing evidence, provider errors, or timeouts are evidence of a safe turn.

Relationship to ASLI. The original ASLI Hindi telephone study remains separate. TurnBench does not rewrite that result or borrow its numbers. It provides reusable machinery for future matched turn-taking evaluations in any language.

02 / THE CONTRACT

Three records, one auditable decision.

A score exists only when the recording manifest, adjudicated label, and provider trace agree on a stable decision ID.

1. RECORDING MANIFEST	2. DECISION LABEL	3. PROVIDER TRACE
Stable recording and source IDs, language, condition, target speaker, and source turns.	Pause boundary, outcome, resume or true-end timing, annotation provenance, and final label.	One run ID, provider status or error, and timestamped speech_started, turn_committed, and agent_first_audio events.

LABEL	MEANING	HEADLINE USE
continue	The same target speaker resumes their thought.	PIR only
yield	The target speaker has genuinely yielded the turn.	Response delay only
overlap	Simultaneous speech makes the turn ambiguous.	Excluded, visible in coverage
unclear	Available evidence cannot support a reliable decision.	Excluded, visible in coverage

The anti-shortcut rule

Labels come from the audio and surrounding conversation. A filler word, lexical ending, punctuation, or transcript pattern may be analysed later, but it may never decide whether a turn continued or yielded.

For a real benchmark, TurnBench requires two independent native-language annotations followed by adjudication. Fixture labels are allowed only for examples and tests; they are visibly marked as fixture provenance, not human-study evidence.

03 / THE SCORING

Two measures. Honest denominators.

TurnBench reports interruption and delay separately because a continuation is not a yield.

PREMATURE-INTERRUPTION RATE (PIR)	RESPONSE DELAY
Among scored continue decisions: the share where first audible agent audio occurs before the speaker's continuation resume time. PIR = early replies / scored continuations	For scored yield decisions: first audible agent audio minus the adjudicated true end. The report publishes p50 and p95 in milliseconds. delay = first audio - true end

Coverage is part of the result. The scorer never turns an absence of evidence into a safe answer.

IF THE ROW IS...	TURNBENCH DOES...
overlap or unclear	Excludes it from PIR and delay denominators, but keeps it in explicit coverage counts.
provider failed or timed out	Marks it unavailable and reports its rate. It cannot improve PIR or delay.
agent_first_audio is missing	Marks it unavailable rather than silently treating the agent as safe or fast.

The scorer uses the first **audible** agent audio event. This keeps the metric close to the caller's experience instead of treating a hidden internal decision as the reply.

04 / THE REPORT

Comparable across languages without hiding the weak spots.

The report keeps a multilingual headline and an operational pooled view, but does not confuse one with the other.

VIEW	HOW IT AGGREGATES	WHY IT EXISTS
overall	Macro by language: equal mean of defined per-language values.	A large language subset cannot erase a weak smaller language.
micro_overall	Micro by decision: pooled eligible decisions.	Shows the operational, decision-weighted view.
grouped rows	Micro summaries by language, condition, and source recording.	Makes it possible to locate where behaviour changes.

Provenance

run_id, provider, config
and label provenance

Counts

actual decision, trace
and source-recording counts

Uncertainty

1,000 source-group draws
for micro PIR only

No pretend precision

A language with no eligible continuation does not receive a zero PIR. A language with no eligible yield does not receive a zero delay. If fewer than two eligible source recordings exist, the source-group bootstrap bounds are null rather than invented.

All input links are validated before a report is emitted: unique IDs, matching recording and trace provenance, one shared provider run ID, coherent timestamps, and standard JSON without duplicate keys. Multilingual JSONL reads and writes are UTF-8.

05 / USING IT

Run the local fixture suite in one command.

The current command is intentionally small: it scores local, versioned inputs and writes one report. It does not call a provider or need a key.

```
uv run asli-turnbench score \  
--recordings turnbench/fixtures/recordings.jsonl \  
--labels turnbench/fixtures/labels.jsonl \  
--events turnbench/fixtures/events.jsonl \  
--out /tmp/turnbench-report.json \  
--provider fixture
```

What has been built in v1

DONE	DETAIL
Separate CLI	asli-turnbench is distinct from the existing ASLI CLI and study workflow.
Versioned records	Recording, label, trace, and report schemas with strict cross-record validation.
Timing resolver	Mechanical decision boundaries and event ordering checks protect metric semantics.
Score + report	PIR, response delay, coverage, macro and micro summaries, and source-group uncertainty.
Fixtures + tests	Offline fixtures, annotation guidance, and 176 passing project checks at release.

What must happen before external claims. Gather matched real recordings, annotate each decision independently in the speaker's language, replay the exact provider configuration through a separate adapter, and publish the rows plus availability. Only then should results be compared across languages, conditions, or providers.

TurnBench is ready to make a later study reproducible. It is deliberately not pretending that the fixtures are a live benchmark result.

Repository references: docs/turnbench.md, turnbench/ANNOTATION.md, and the asli.turnbench package.
Implementation commit: c45b40e.